

■ SEIR-I Lab 2 — GCP Terraform Deployment

■ Iowa VM + Startup Script + HTTP Service on Port 80

■ Goal	Deploy a Google Compute Engine VM in us-central1-a with Terraform, allow HTTP traffic on port 80, and use a startup script to install nginx and serve the ops panel endpoints / , /healthz , and /metadata .
■ Workforce relevance	This lab demonstrates the shift from click-ops to cloud engineering through reproducible deployments, version-controlled infrastructure, predictable changes, and reviewable diffs.

■ Deliverables included in this report

- A polished project overview and execution plan.
- Expanded introduction and conclusion for submission-ready wording.
- Terraform and startup-script code snippets for technical proof.
- Validation steps for the gate script.
- Screenshots from the apply output and successful webpage.

■ *Final thought: You didn't click your way here — you automated.*

■ Project Outline

1) Core objectives

- Provision the Lab 1 VM stack with Terraform.
- Create a firewall rule to allow HTTP on port 80.
- Use a startup script to install nginx and serve the ops panel.
- Keep the build in Iowa by default: region us-central1, zone us-central1-a.

2) Workflow plan

- Initialize Terraform and provider dependencies.
- Validate syntax and references before deployment.
- Generate a saved plan and review the proposed infrastructure changes.
- Apply the plan, retrieve the VM URL, and open the service in a browser.
- Run the gate script against the VM external IP.

■ 1. Project Objective

The purpose of this project was to build a repeatable Google Cloud deployment with Terraform instead of configuring infrastructure manually. The VM is expected to expose a custom web page on `/`, a basic health endpoint on `/healthz`, and a metadata JSON endpoint on `/metadata`. This turns the environment into something testable, observable, and easy to reproduce for future changes.

■ 2. Workforce Relevance

- ■ Reproducible deployments — the same code can deploy the same infrastructure again.
- ■ Version-controlled infrastructure — Terraform files can be tracked, reviewed, and improved over time.
- ■ Predictable changes — terraform plan shows what will happen before apply.
- ■ Infrastructure as Code — the environment becomes documentation and automation at the same time.

■ 3. Architecture Overview

Component	Purpose
■ Compute Engine VM	Hosts the Linux instance and the nginx web service.
■ Firewall Rule	Allows inbound TCP traffic on port 80 for HTTP access.
■ Startup Script	Installs packages, collects metadata, writes files, and restarts nginx.
■ Region / Zone	Uses us-central1 and us-central1-a by default for the Iowa deployment.

■ 4. Terraform Workflow

The project folder should contain the Terraform configuration files, the service-account JSON file, and the HTTP gate script. The standard command sequence for the lab is shown below.

```
terraform init
terraform validate
terraform plan -out tfplan
terraform apply tfplan
terraform output vm_url
```

■ 5. Startup Script Responsibilities

- ■ Install required packages: nginx, curl, and jq.
- ■ Pull instance and network information from the GCP metadata server.
- ■ Generate **index.html** and **metadata.json** inside `/var/www/html`.
- ■ Expose simple test endpoints for operational validation.
- ■ Test the nginx configuration and restart the service cleanly.

■ 6. Key Code Snippets

Terraform / CLI workflow

```
terraform init
terraform validate
terraform plan -out tfplan
terraform apply tfplan
terraform output vm_url
```

Startup script core logic

```
#!/bin/bash
set -euo pipefail
exec > >(tee /var/log/startup-script.log) 2>&1
set -x

apt-get update -y
apt-get install -y nginx curl jq
mkdir -p /var/www/html

METADATA="http://metadata.google.internal/computeMetadata/v1"
HDR="Metadata-Flavor: Google"
md() { curl -fsS -H "$HDR" "$METADATA/$1" || echo "unknown"; }

nginx -t
systemctl enable nginx >/dev/null 2>&1 || true
systemctl restart nginx
```

Validation gate command

```
VM_IP=$(terraform output -raw vm_external_ip)
VM_IP="$VM_IP" ./gate_lab2_http.sh
```

■ 7. Web Endpoints

Endpoint	Expected Result	Why It Matters
/	HTTP 200	Confirms nginx is serving the landing page.
/healthz	ok	Provides a fast health check for the service.
/metadata	Valid JSON	Exposes infrastructure details that the gate script verifies.

■ 8. Validation and Gate Test

The gate script is the final pass/fail check for the project. It verifies that the root endpoint returns 200, that **/healthz** returns **ok**, and that **/metadata** is valid JSON containing the required network fields.

■ Gate requirements

- / returns HTTP 200.
- /healthz returns exactly ok.
- /metadata returns valid JSON.
- The JSON includes region, network.vpc, and network.subnet.

```
VM_IP=$(terraform output -raw vm_external_ip)
VM_IP="$VM_IP" ./gate_lab2_http.sh
```

■■ 9. Troubleshooting and Lessons Learned

- ■■ Terraform vs. Bash interpolation required care because \${...} inside heredocs can be parsed by Terraform.
- ■ Startup script changes sometimes needed VM recreation so the latest script would actually run on boot.
- ■ Seeing the default nginx page meant nginx was alive, but the custom site content or config had not fully taken over.
- ■ Logging was essential for debugging startup behavior and seeing where the script stopped.
- ■ Creating /var/www/html explicitly helped avoid file-write assumptions during boot.
- ■ Running nginx -t before restart made service failures easier to catch and explain.

■ 10. Screenshots / Proof of Work

The report includes proof that the Terraform deployment executed and that the webpage was reached successfully. These screenshots support the project narrative and can be used in the final submission

packet.

■ Terraform apply / deployment proof

```
Enter a value: yes
google_compute_instance.traveler_vm: Destroying... [id=projects/whannah-gcp-75/zones/us-central1-a/instances/traveler-node-lab2]
google_compute_instance.traveler_vm: Still destroying... [id=projects/whannah-gcp-75/zones/us-central1-a/instances/traveler-node-lab2, 00m10s elapsed]
google_compute_instance.traveler_vm: Still destroying... [id=projects/whannah-gcp-75/zones/us-central1-a/instances/traveler-node-lab2, 00m20s elapsed]
google_compute_instance.traveler_vm: Still destroying... [id=projects/whannah-gcp-75/zones/us-central1-a/instances/traveler-node-lab2, 00m30s elapsed]
google_compute_instance.traveler_vm: Destruction complete after 31s
terraform_data.startup_script_version: Modifying... [id=db08042c-a4d0-1e85-e576-f5ca1e83fd9f]
terraform_data.startup_script_version: Modifications complete after 0s [id=db08042c-a4d0-1e85-e576-f5ca1e83fd9f]
google_compute_instance.traveler_vm: Creating...
google_compute_instance.traveler_vm: Still creating... [00m10s elapsed]
google_compute_instance.traveler_vm: Still creating... [00m20s elapsed]
google_compute_instance.traveler_vm: Still creating... [00m30s elapsed]
google_compute_instance.traveler_vm: Creation complete after 38s [id=projects/whannah-gcp-75/zones/us-central1-a/instances/traveler-node-lab2]

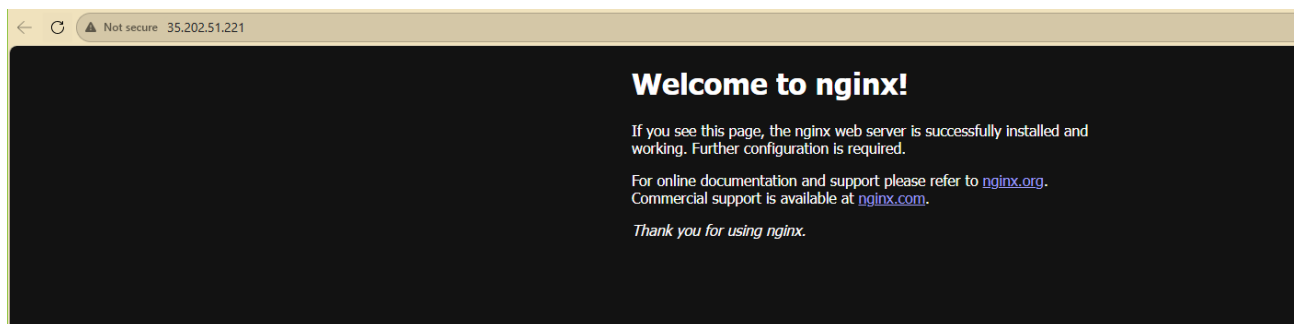
Apply complete! Resources: 1 added, 1 changed, 1 destroyed.

Outputs:

vm_external_ip = "35.202.51.221"
vm_url = "http://35.202.51.221"

Willi_so151lg@Lenovo MINGW64 ~/TheoWAF/class7.5/GCP/Terraform/Week2LabB (Week_2)
$
```

■ Browser proof of successful webpage access



■ 11. Expanded Conclusion

This lab demonstrated more than just launching a VM. It showed how infrastructure can be defined as code, reviewed before deployment, recreated consistently, and validated through automated checks. By combining Terraform, a startup script, nginx, and the gate test, the project reinforced the practical habits of a cloud engineer: automation, repeatability, observability, and structured troubleshooting. In a real workplace setting, these skills support faster deployments, clearer documentation, and more reliable operations.

■ Final checklist

- Terraform initialized, validated, planned, and applied successfully.
- VM URL retrieved and opened in the browser.
- Custom service endpoints designed for validation.

- Gate script workflow documented.

- Technical snippets, screenshots, and polished write-up included in one report.